Most Significant Publications — Tejaswi Venumadhav Nerella

1. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 101, 083030
   Title: New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo

2. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), Physical Review D, 100, 023007
   Title: Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run

3. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009
   Title: New binary black hole mergers in the LIGO-Virgo O3a data

4. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2019), Physical Review D, 100, 023011
   Title: New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO

5. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030
   Title: Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers

Refereed Publications

1. Hegade K. R., A., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N., (2026), Physical Review Letters, 136, 071401
   Title: Relativistic and Dynamical Love Numbers

2. Cheung, M.H.Y., Wadekar, D., Mehta, A.K., Islam, T., Roulet, J., Berti, E., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2026), Physical Review D, 113, 023003
   Title: Searching for intermediate mass ratio binary black hole mergers in the third observing run of LIGO-Virgo-KAGRA

3. Mehta, A.K., Wadekar, D., Anantpurkar, I., Roulet, J., **Venumadhav, T.**, Islam, T., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 124023
   Title: Binary black hole population inference combining confident and marginal events from the IAS-HM search pipeline

4. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 112, 104039
   Title: Post-Newtonian theory-inspired framework for characterizing eccentricity in gravitational waveforms

5. Perera, A., Zackay, B., **Venumadhav, T.**, (2025), Astrophys. J. Suppl. Ser., 281, 4
   Title: A New Search Pipeline for Short Gamma-Ray Bursts in Fermi/GBM Data—A 50% Increase in the Number of Detections

6. Mushkin, J., Roulet, J., Zackay, B., **Venumadhav, T.**, Ivashtenko, O., Wadekar, D., Mehta, A.K., Zaldarriaga, M., (2025), Physical Review D, 112, 104025
   Title: Sampler-free gravitational wave inference using matrix multiplication

7. Jana, S., Kapadia, S.J., **Venumadhav, T.**, More, S., Ajith, P., (2025), Physical Review Letters, 135, 111402
   Title: Probing the Nature of Dark Matter Using Strongly Lensed Gravitational Waves from Binary Black Holes

8. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 044070
   Title: Data-driven extraction, phenomenology, and modeling of eccentric harmonics in binary black hole merger waveforms

9. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 111, L081503
   Title: Universal phenomenological relations between spherical harmonic modes in nonprecessing eccentric binary black hole merger waveforms

10. Mehta, A.K., Olsen, S., Wadekar, D., Roulet, J., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 111, 024049
    Title: New binary black hole mergers in the LIGO-Virgo O3b data

11. Jana, S., J Kapadia, S., **Venumadhav, T.**, More, S., Ajith, P., (2024), Classical and Quantum Gravity, 41, 245010
    Title: Strong-lensing cosmography using third-generation gravitational-wave detectors

12. Wadekar, D., **Venumadhav, T.**, Mehta, A.K., Roulet, J., Olsen, S., Mushkin, J., Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 084035
    Title: New approach to template banks of gravitational waves with higher harmonics: Reducing matched-filtering cost by over an order of magnitude

13. Chia, H.S., Edwards, T.D.P., Wadekar, D., Zimmerman, A., Olsen, S., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 063007
    Title: In pursuit of Love numbers: First templated search for compact objects with large tidal deformabilities in the LIGO-Virgo data

14. Wadekar, D., **Venumadhav, T.**, Roulet, J., Mehta, A.K., Zackay, B., Mushkin, J., Zaldarriaga, M., (2024), Physical Review D, 110, 044063
    Title: New search pipeline for gravitational waves with higher-order modes using mode-by-mode filtering

15. Roulet, J., Mushkin, J., Wadekar, D., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 044010
    Title: Fast marginalization algorithm for optimizing gravitational wave detection, parameter estimation, and sky localization

16. Roulet, J., **Venumadhav, T.**, (2024), Annu. Rev. Nucl. Part. Sci., 74, 207
    Title: Inferring Binary Properties from Gravitational-Wave Signals

17. Yu, H., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2023), Physical Review D, 108, 064059
    Title: Accurate and efficient waveform model for precessing binary black holes

18. Jana, S., Kapadia, S.J., **Venumadhav, T.**, Ajith, P., (2023), Physical Review Letters, 130, 261401
    Title: Cosmography Using Strongly Lensed Gravitational Waves from Binary Black Holes

19. Yu, H., Weinberg, N.N., Arras, P., Kwon, J., **Venumadhav, T.**, (2023), Mon. Not. R. Astron. Soc., 519, 4325
    Title: Beyond the linear tide: impact of the non-linear tidal response of neutron stars on gravitational waveforms from binary inspirals

20. Roulet, J., Olsen, S., Mushkin, J., Islam, T., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 123015
    Title: Removing degeneracy and multimodality in gravitational wave source parameters

21. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009
Title: New binary black hole mergers in the LIGO-Virgo O3a data

22. Chia, H.S., Olsen, S., Roulet, J., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 024009
Title: Signs of higher multipoles and orbital precession in GW151226

23. Olsen, S., Roulet, J., Chia, H.S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083036
Title: Mapping the likelihood of GW190521 with diverse mass and spin priors

24. Roulet, J., Chia, H.S., Olsen, S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083010
Title: Distribution of effective spins and masses of binary black holes from the LIGO and Virgo O1-O3a observing runs

25. Zackay, B., **Venumadhav, T.**, Roulet, J., Dai, L., Zaldarriaga, M., (2021), Physical Review D, 104, 063034
Title: Detecting gravitational waves in data with non-stationary and non-Gaussian noise

26. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030
Title: Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers

27. Roulet, J., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 102, 123022
Title: Binary black hole mergers from LIGO/Virgo O1 and O2: Population inference combining confident and marginal events

28. Huang, Y., Haster, C.J., Roulet, J., Vitale, S., Zimmerman, A., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 102, 103024
Title: Source properties of the lowest signal-to-noise-ratio binary black hole detections

29. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 101, 083030
Title: New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo

30. Dai, L., Kaurov, A.A., Sharon, K., Florian, M., Miralda-Escudé, J., **Venumadhav, T.**, Frye, B., Rigby, J.R., Bayliss, M., (2020), Mon. Not. R. Astron. Soc., 495, 3192
Title: Asymmetric surface brightness structure of caustic crossing arc in SDSS J1226+2152: a case for dark matter substructure

31. Samsing, J., **Venumadhav, T.**, Dai, L., Martinez, I., Batta, A., Lopez, M., Ramirez-Ruiz, E., Kremer, K., (2019), Physical Review D, 100, 043009
Title: Probing the black hole merger history in clusters using stellar tidal disruptions

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Title: Highly Magnified Stars in Lensing Clusters: New Evidence in a Galaxy Lensed by MACS J0416.1-2403

33. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2019), Physical Review D, 100, 023011
Title: New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO

34. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), Physical Review D, 100, 023007
Title: Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run
35. Roulet, J., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2019), Physical Review D, 99, 123022
Title: Template bank for compact binary coalescence searches in gravitational wave data: A general geometric placement algorithm
36. **Venumadhav, T.**, Dai, L., Kaurov, A., Zaldarriaga, M., (2018), Physical Review D, 98, 103513
Title: Heating of the intergalactic medium by the cosmic microwave background during cosmic dawn
37. Dai, L., **Venumadhav, T.**, Kaurov, A.A., Miralda-Escud, J., (2018), Astrophysical Journal, 867, 24
Title: Probing Dark Matter Subhalos in Galaxy Clusters Using Highly Magnified Stars
38. Kaurov, A.A., **Venumadhav, T.**, Dai, L., Zaldarriaga, M., (2018), Astrophysical Journal, 864, L15
Title: Implication of the Shape of the EDGES Signal for the 21 cm Power Spectrum
39. Hirata, C.M., Mishra, A., **Venumadhav, T.**, (2018), Physical Review D, 97, 103521
Title: Detecting primordial gravitational waves with circular polarization of the redshifted 21 cm line. I. Formalism
40. **Venumadhav, T.**, Dai, L., Miralda-Escudé, J., (2017), Astrophysical Journal, 850, 49
Title: Microlensing of Extremely Magnified Stars near Caustics of Galaxy Clusters
41. Gluscevic, V., **Venumadhav, T.**, Fang, X., Hirata, C., Oklopčić, A., Mishra, A., (2017), Physical Review D, 95, 083011
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42. **Venumadhav, T.**, Oklopčić, A., Gluscevic, V., Mishra, A., Hirata, C.M., (2017), Physical Review D, 95, 083010
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43. Dai, L., **Venumadhav, T.**, Sigurdson, K., (2017), Physical Review D, 95, 044011
Title: Effect of lensing magnification on the apparent distribution of black hole mergers
44. **Venumadhav, T.**, Chang, T.C., Doré, O., Hirata, C.M., (2016), Astrophysical Journal, 826, 116
Title: A Practical Theorem on Using Interferometry to Measure the Global 21-cm Signal
45. **Venumadhav, T.**, Cyr-Racine, F.Y., Abazajian, K.N., Hirata, C.M., (2016), Physical Review D, 94, 043515
Title: Sterile neutrino dark matter: Weak interactions in the strong coupling epoch
46. **Venumadhav, T.**, Hirata, C., (2015), Physical Review D, 91, 123009
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47. **Venumadhav, T.**, Zimmerman, A., Hirata, C.M., (2014), Astrophysical Journal, 781, 23
Title: The Stability of Tidally Deformed Neutron Stars to Three- and Four-mode Coupling
48. Venumadhav, T., Haque, M., Moessner, R., (2010), Physical Review B, 81, 054305
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1. R., A.H.K., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N., (2026), arXiv:2605.08569
Title: The Good, the Bad, and the Subtle: Relativistic mode sums for neutron-star tidal response

2. Islam, T., Wadekar, D., **Venumadhav, T.**, Zaldarriaga, M., Mehta, A.K., Roulet, J., Zackay, B., (2026), arXiv:2605.11280
Title: Discovery of Interpretable Surrogates via Agentic AI: Application to Gravitational Waves
3. Islam, T., **Venumadhav, T.**, Wadekar, D., Mehta, A.K., Roulet, J., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2604.07388
Title: GW190711_030756 and GW200114_020818: astrophysical interpretation of two asymmetric binary black hole mergers in the IAS catalog
4. Zhou, Z., Wadekar, D., Roulet, J., Ivashtenko, O., **Venumadhav, T.**, Islam, T., Mehta, A.K., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2603.05784
Title: Searching for precessing binary systems with mode-by-mode filtering and marginalization
5. Islam, T., **Venumadhav, T.**, Wadekar, D., (2026), arXiv:2601.18986
Title: Progenitor of the recoiling super-massive black hole RBH-1 identified using HST/JWST imaging
6. Maity, K.N., Jana, S., **Venumadhav, T.**, Barsode, A., Ajith, P., (2025), arXiv:2512.15168
Title: Strong lensing cosmography using binary-black-hole mergers: Prospects for the near future
7. Islam, T., **Venumadhav, T.**, Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), arXiv:2509.20556
Title: Data-driven extraction and phenomenology of eccentric harmonics in eccentric spinning binary black hole mergers
8. Wadekar, D., Pimpalkar, A., Ho-Yeuk Cheung, M., Wandelt, B., Berti, E., Mehta, A.K., **Venumadhav, T.**, Roulet, J., Islam, T., Zackay, B., Mushkin, J., Zaldarriaga, M., (2025), arXiv:2507.08318
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9. Schiff, J., **Venumadhav, T.**, (2025), arXiv:2506.16517
Title: Primordial magnetic fields and modified recombination histories
10. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), arXiv:2504.12420
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11. Kwon, K.J., Yu, H., **Venumadhav, T.**, (2025), arXiv:2503.11837
Title: Resonance locking: radian-level phase shifts due to nonlinear hydrodynamics of g-modes in merging neutron star binaries
12. Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., Islam, T., (2025), arXiv:2501.17939
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13. Kwon, K.J., Yu, H., **Venumadhav, T.**, (2024), arXiv:2410.03831
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15. Islam, T., Roulet, J., **Venumadhav, T.**, (2022), arXiv:2210.16278
Title: Factorized Parameter Estimation for Real-Time Gravitational Wave Inference

16. Dai, L., Zackay, B., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2020), arXiv:2007.12709
Title: Search for Lensed Gravitational Waves Including Morse Phase Information: An Intriguing Candidate in O2
17. Coleman, M.S.B., **Venumadhav, T.**, Zackay, B., (2019), arXiv:1903.04978
Title: Gravitational-wave-moderated Accretion: The Case of ES Ceti
18. Haris, K., Mehta, A.K., Kumar, S., **Venumadhav, T.**, Ajith, P., (2018), arXiv:1807.07062
Title: Identifying strongly lensed gravitational wave signals from binary black hole mergers
19. Zackay, B., Dai, L., **Venumadhav, T.**, (2018), arXiv:1806.08792
Title: Relative Binning and Fast Likelihood Evaluation for Gravitational Wave Parameter Estimation
20. Dai, L., **Venumadhav, T.**, Zackay, B., (2018), arXiv:1806.08793
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21. Dai, L., **Venumadhav, T.**, (2017), arXiv:1702.04724
Title: On the waveforms of gravitationally lensed gravitational waves

n^{th} Author Papers

1. Raaijmakers, G., et. al., (2021), Astrophysical Journal, 922, 269
Title: The Challenges Ahead for Multimessenger Analyses of Gravitational Waves and Kilonova: A Case Study on GW190425
2. Weltman, A., et. al., (2020), Publ. Astron. Soc. Aust., 37, e002
Title: Fundamental physics with the Square Kilometre Array
3. Doré, O., et. al., (2014), arXiv:1412.4872
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